plant biology

Plant Biology ISSN 1435-8603

RESEARCH PAPER

Effects of clonal fragmentation on intraspecific competition of a stoloniferous floating plant

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Keywords

Aquatic plant; clonal plant; log response ratio; physiological integration; *Pistia stratiotes*; size–number relationships.

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Editor

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Received: 3 December 2013;
Accepted: 26 January 2014

doi:10.1111/plb.12170

ABSTRACT

Disturbance is common and can fragment clones of plants. Clonal fragmentation may affect the density and growth of ramets so that it could alter intraspecific competition. To test this hypothesis, we grew one (low density), five (medium density) or nine (high density) parent ramets of the floating invasive plant *Pistia stratiotes* in buckets, and newly produced offspring ramets were either severed (with fragmentation) or remained connected to parent ramets (no fragmentation). Increasing density reduced biomass of the whole clone (*i.e.* parent ramet plus its offspring ramets), showing intense intraspecific competition. Fragmentation decreased biomass of offspring ramets, but increased biomass of parent ramets and the whole clone, suggesting significant resource translocation from parent to offspring ramets when clones were not fragmented. There was no interaction effect of density x fragmentation on biomass of the whole clone, and fragmentation did not affect competition intensity index. We conclude that clonal fragmentation does not alter intraspecific competition between clones of *P. stratiotes*, but increases biomass production of the whole clone. Thus, fragmentation may contribute to its interspecific competitive ability and invasiveness, and intentional fragmentation should not be recommended as a measure to stop the rapid growth of this invasive species.

INTRODUCTION

Disturbance is ubiquitous not only in terrestrial ecosystems but also in aquatic ones (Klimešová & Klimeš 2003; Klimešová *et al.* 2008; Puijalon *et al.* 2008; Latzel & Klimešová 2009). In aquatic ecosystems, strong currents, flooding, grazing and trampling by waterfowl and physical damage through transportation and fishing vessels are different types of disturbance commonly encountered by aquatic plants (Barrat-Segretain & Amoros 1995). Such disturbance can break connected clones of aquatic plants into fragments of different sizes (Sainty *et al.* 1998; Stuefer & Huber 1999). Although effects of clonal fragmentation on the growth of clonal plants have been addressed (Birch & Hutchings 1999; Stuefer & Huber 1999; Dong *et al.* 2012), little is known about effects of clonal fragmentation on intraspecific competition of aquatic plants.

Floating hydrophytes such as *Eichhornia crassipes* and *Pistia stratiotes* can produce asexual individuals (ramets) connected through stolons (Richards 1982; Neuenschwander *et al.* 2009). Due to little lignification of the tissues (Barrat-Segretain 1996), stolons can be easily broken by disturbance so that clones formerly consisting of interconnected ramets become fragmented. When a clone is intact, connected ramets can exchange resources such as water, carbohydrates and nutrients (Alpert 1996; Xu *et al.* 2010, 2012a,b) and possibly also defence signals (Gómez *et al.* 2008; Chen *et al.* 2011; Jelínková *et al.* 2012) through physiological integration so that the growth of individual ramets and competition among ramets of the same clones can be regulated (Holzapfel & Alpert 2003; Rivera & Scrosati 2008; Roiloa & Retuerto 2012). Furthermore, inter-ramet connections may also promote self/non-self recognition, which can further reduce intraspecific competition (Holzapfel & Alpert 2003; Gruntman & Novoplansky 2004; Roiloa *et al.* 2013).

On the other hand, after fragmentation ramets will not continue to exchange resources and thus will depend on each other. In vast water areas (*e.g.* large rivers, streams or lakes), fragmentation of clones is believed to increase clonal dispersal because ramets can move along with the water current (Okada *et al.* 2009; Dong *et al.* 2012). However, in a particular limited space (*e.g.* small ponds and ditches), fragmentation may greatly increase the density of isolated, independent individuals (Dong *et al.* 2012), and reduce the possibility of clone-level regulation (*e.g.* self/non-self recognition; Holzapfel & Alpert 2003; Gruntman & Novoplansky 2004; Roiloa *et al.* 2013). Furthermore, fragmentation may greatly decrease the growth of whole clones (Roiloa & Retuerto 2006; Roiloa *et al.* 2010). These effects of fragmentation may change when the density of the initial clones increases and thus fragmentation may greatly affect intraspecific competition. So far, however, we still know very little about how clonal fragmentation affects the growth and intraspecific competition of floating plants.

We grew one (low density), five (medium density) or nine (high density) parent ramets of the floating invasive plant *Pistia stratiotes* in buckets, and all newly produced offspring ramets were either severed (with fragmentation) or remained connected to the parent ramets (no fragmentation). Specifically, we address the following questions: (i) does initial density affect the growth of clones of *P. stratiotes*? We predicted that

growth of the clones would decrease with increasing density. (ii) Does clonal fragmentation affect growth of the clones? We predicted that clonal fragmentation would decrease growth of the clones. (iii) Do the effects of fragmentation change with increasing clone density so that clonal fragmentation affects the intraspecific competition between clones? We predicted that the effects of fragmentation would change with increasing clone density and fragmentation would alter the intensity of intraspecific competition between clones.

MATERIAL AND METHODS

The species

Pistia stratiotes L. (Araceae), commonly known as water lettuce, is a free-floating, stoloniferous, perennial aquatic weed (Wu & Raven 2003). It originates from South America and is now widely distributed in tropical and subtropical regions throughout the world (Wu & Raven 2003; Neuenschwander *et al.* 2009). *Pistia stratiotes* can propagate sexually by producing seeds and asexually by forming offspring ramets connected through stolons (Neuenschwander *et al.* 2009). Each ramet consists of a dense rosette of overlapping leaves with short hairs, a very short and compressed stem and feathery roots (Neuenschwander *et al.* 2009).

Pistia stratiotes is listed as an invasive species in many countries, including China (Li & Xie 2002; Neuenschwander *et al.* 2009). It is commonly found in lakes, rivers, ponds, ditches or other slow-moving or stagnant water bodies, and can have a serious effect on local ecosystem processes (Howard & Harley 1998; Neuenschwander *et al.* 2009). It relies mainly on vegetative means to spread, occupying the whole water surface, blocking channels and interfering with fisheries. The dense mats formed by *P. stratiotes* also decrease light penetration and oxygen content in water, thereby negatively affecting productivity and diversity of local species (Howard & Harley 1998).

Experimental material

In July 2011, plants of *P. stratiotes* were obtained from Beijing Da Senlin Flower Market and 230 plants of similar size were selected. These plants might or might not differ in genotype, and were randomly assigned to the treatments described below to avoid potential confounding effects of genotype. Each plant (parent ramet) had 14–17 leaves, and its stolons and offspring ramets (if present) were removed. Then 20 parent ramets were randomly selected for measuring initial biomass (0.392 ± 0.175 g, mean ± SE) and the remaining 210 were used for the experiment described below. The crown diameter of the parent ramets was 5–7 cm.

Experimental design

The experiment used a factorial design with three levels of initial plant density (low, medium or high) and two levels of fragmentation (with or without fragmentation). In the low-density treatment, one parent ramet of *P. stratiotes* was planted in a bucket (37-cm diameter and 45-cm height), and in the medium- and high-density treatments five and nine parent ramets were planted, respectively. The bottom of each bucket was filled with a 15-cm deep riparian soil (0.80 g total N, 0.59 g total P and 13.20 g total organic matter·kg^{-1} soil), and then 25-cm deep tap water (11.20 mg total N and 9.19 mg total P·l^{-1} tap water) added above the soil surface. New stolons and offspring ramets of *P. stratiotes* started to emerge after 2 days. In half of the buckets in each density treatment, we severed stolon connections between the parent ramets and the newly produced offspring ramets, and in the other half the stolons were left intact. We severed the middle of stolons only when the offspring ramets had roots and the length of stolons was 3–5 cm. The offspring ramets had three to four leaves when fragmented. During the experiment, isolated offspring ramets did not produce any granddaughter ramets, and thus all fragments in the severing treatment contained only one offspring ramet. The severing was repeated every other day during the experiment. Each treatment had seven replicates.

The experiment lasted for 6 weeks from 8 August 2011 to 17 September 2011, and was conducted in an open area close to the Miyun Reservoir (40°55′ N, 117°01′ E) in Beijing. Water in buckets was replenished every 2 weeks. The mean temperature in the water (20 cm below the water surface in the buckets) was 23.2 ± 0.6 °C (mean ± SE).

Harvest and measurements

During the experiment, every other day we counted the number of individual ramets after severing. At harvest, we counted the total number of individual offspring ramets and total number of leaves of offspring ramets in each bucket. We also counted the number of surviving parent ramets and total number of leaves of the surviving parent ramets. Parent ramets were much larger than offspring ramets and thus very easy to distinguish. Parent and offspring ramets were oven-dried at 70 °C for 48 h and weighed.

Data analysis

All parent ramets survived. Before analysis, values of all variables were divided by the number of parent ramets (*i.e.* 1, 5 or 9), so that the values were scaled to the level of per parent ramet. All data, except biomass of the whole clone (*i.e.* parent ramet plus its offspring ramets) and number of offspring ramets, were log-transformed to increase homogeneity of variance. We used two-way ANOVA to test the effects of density and fragmentation on biomass and number of leaves of the whole clone, of the parent ramets and of the offspring ramets.

To measure intraspecific interaction intensity, we calculated the log response ratio (LnRR) as $\mathrm{LnRR} = \ln(B_i/B_0)$, where B_i is the biomass of plants in the high- or medium-density treatments for replicate i and B_0 is the mean biomass of plants over the seven replicates in the low-density treatment. Values of LnRR are symmetrical around zero, with negative values indicating competition and positive values indicating facilitation (James & Richards 2007). We used two-way ANOVA to test the effect of density (medium *versus* high) and fragmentation on LnRR.

We calculated biomass and number of leaves per offspring ramet as measures of the mean size of offspring ramets. We used two-way ANOVA to test the effects of density and fragmentation on the size of offspring ramets, and linear regressions to examine relationships between the size and number of offspring ramets with and without fragmentation. The software package SPSS 16.0 (SPSS, Chicago, IL, USA) was used for all analyses.

RESULTS

Effects of density and fragmentation at the clone level

Increasing density significantly decreased biomass and number of leaves of the whole clone (*i.e.* the parent ramet plus its offspring ramets) of *P. stratiotes* (Table 1A, Fig. 1), and log response ratio (LnRR) was significantly smaller in the high- than in the low-density treatment (Fig. 2, effect of density: $F_{1,24} = 4.40$, $P = 0.047$). Fragmentation markedly increased biomass of the whole clone but did not affect number of leaves (Table 1A, Fig. 1). There was no significant interaction effect of density x fragmentation on biomass or number of leaves (Table 1A, Fig. 1), and no significant effect of fragmentation ($F_{1,24} = 0.08$, $P = 0.783$) or interaction effect of fragmentation x density ($F_{1,24} = 0.75$, $P = 0.394$) on LnRR (Fig. 2).

Effects of density and fragmentation at parent and offspring ramet levels

Compared with low density (no competition), medium and high density increased biomass and number of leaves of parent ramets (Table 1B, Fig. 3). Fragmentation significantly increased biomass and number of leaves of parent ramets and such effects decreased with increasing density (Table 1B, Fig. 3).

Increasing density significantly decreased the number of offspring ramets, total biomass and total number of leaves of offspring ramets (Table 1C, Fig. 4A–C). Fragmentation significantly decreased total biomass of offspring ramets but did not affect the number of offspring ramets or total number of leaves of offspring ramets (Table 1C, Fig. 4A–C). There was no significant interaction effect of density x fragmentation on the number of offspring ramets, total biomass or total number of leaves of offspring ramets (Table 1C, Fig. 4A–C, Fig. S1). Fragmentation significantly decreased biomass and number of leaves per offspring ramet, and such effects were higher in the medium- and high-density treatments than in the low-density treatment (Table 1C, Fig. 4D and E).

Table 1. ANOVAS for effects of density and fragmentation on the growth of (A) the whole clone, (B) parent ramets and (C) offspring ramets of *Pistia stratiotes*. Values of $P = 0.05$–0.01 are in italic, those of $P < 0.01$ are in bold.

	Density		Fragmentation		Interaction	
Effect	$F_{2, 36}$	P	$F_{1, 36}$	P	$F_{2, 36}$	P
(A) Clone						
Biomass	**13.85**	**<0.001**	*7.04*	*0.012*	0.63	0.540
Number of leaves	**54.3**	**<0.001**	0.67	0.561	0.58	0.749
(B) Parent ramets						
Biomass	**11.00**	**<0.001**	**42.17**	**<0.001**	**10.36**	**<0.001**
Number of leaves	**5.34**	**0.009**	**16.59**	**<0.001**	**5.66**	**0.007**
(C) Offspring ramets						
Biomass	**51.94**	**<0.001**	**44.59**	**<0.001**	2.99	0.110
Number of leaves	**63.85**	**<0.001**	2.20	0.147	0.55	0.582
Number of ramets	**41.24**	**<0.001**	0.11	0.737	0.77	0.469
Biomass per ramet	**5.84**	**0.006**	**78.97**	**<0.001**	**7.56**	**0.002**
Leaf number per ramet	0.63	0.540	**48.96**	**<0.001**	**8.30**	**0.001**

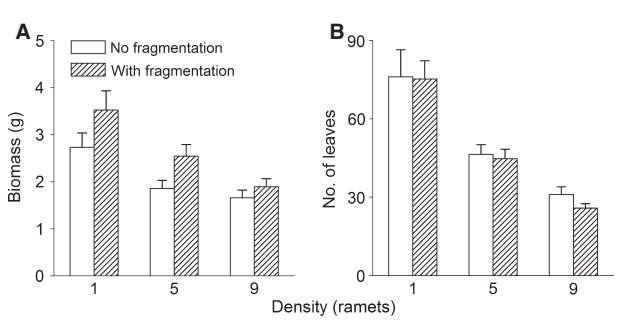

Fig. 1. Effects of density and fragmentation on growth of the whole clone of *Pistia stratiotes*. Mean + SE of (A) biomass and (B) leaf number.

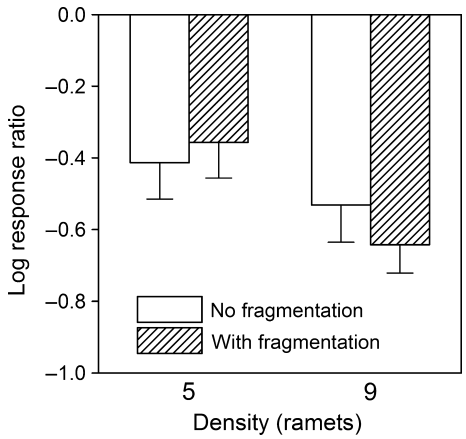

Fig. 2. Effects of density and fragmentation on log response ratio of the whole clone of *Pistia stratiotes*. Bars are means ± SE.

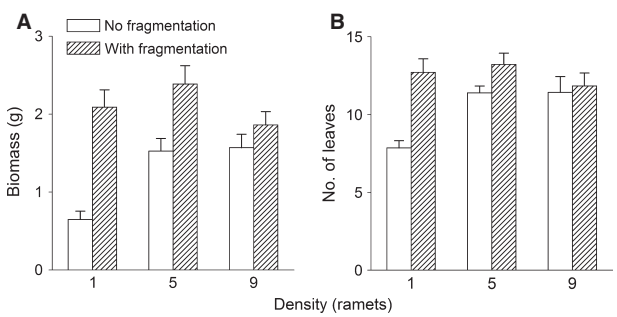

Fig. 3. Effects of density and fragmentation on parent ramets of *Pistia stratiotes*. Mean + SE of (A) biomass and (B) leaf number.

Relationships between the number and size of offspring ramets

Biomass and number of leaves per offspring ramet were significantly negatively correlated with the number of offspring ramets when clones were not fragmented, but were significantly positively correlated with the number of offspring ramets when clones were fragmented (Fig. 5).

DISCUSSION

As expected, increasing density decreased biomass of the whole clone (*i.e.* parent ramet plus its offspring ramets); it also decreased total biomass of offspring ramets. These results suggest that there was intense intraspecific competition

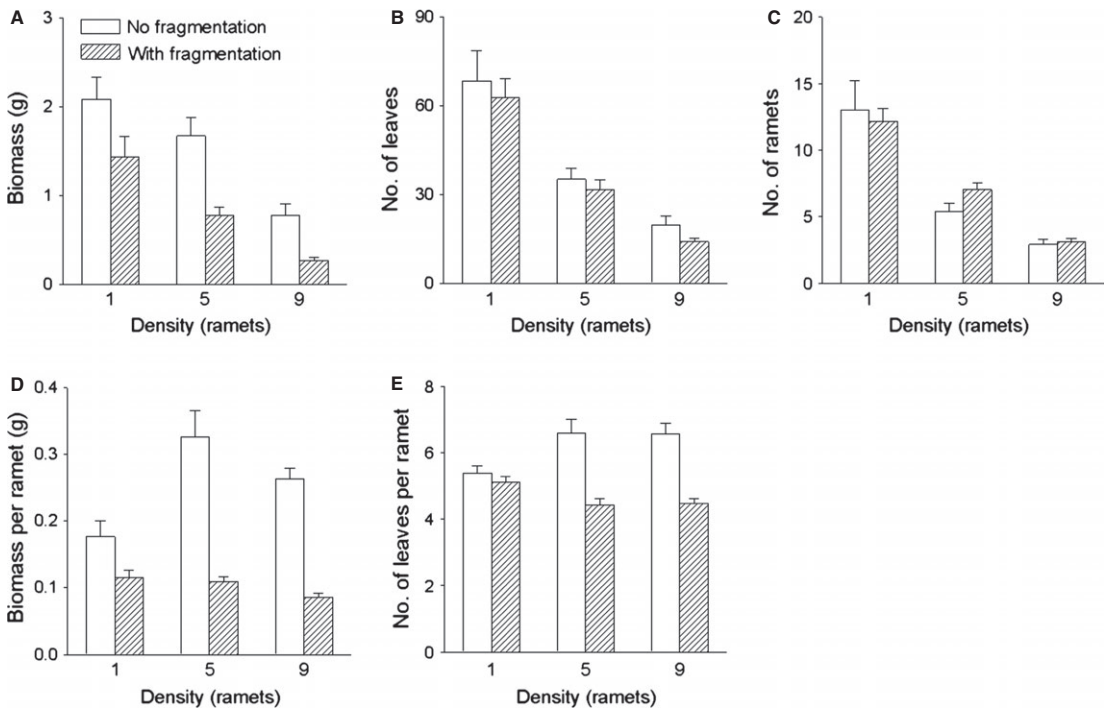

Fig. 4. Effects of density and fragmentation on offspring ramets of *Pistia stratiotes*. Mean + SE of (A) biomass, (B) leaf number, (C) number of ramets, (D) biomass per ramet and (E) leaf number per ramet.

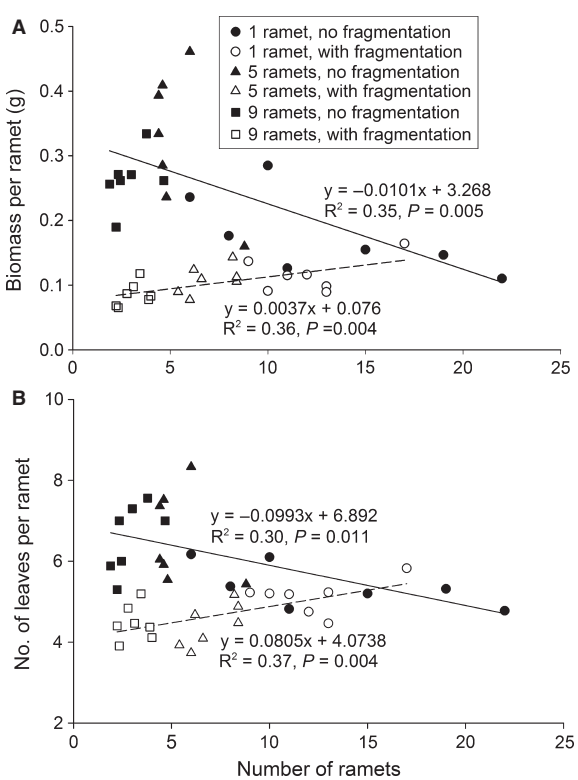

Fig. 5. Effects of fragmentation on the relationships between size, as measured by (A) biomass per ramet and (B) number of leaves per ramet, and number of offspring ramets of *Pistia stratiotes*.

between the clones of *P. stratiotes*, which is in line with the size-dependent theory of intraspecific competition (Bonser & Reader 1995; Tiffin 2002; Bownes *et al.* 2010).

Effects of fragmentation on growth

Studies generally show that clonal fragmentation reduces the growth of recipient ramets (offspring, younger or stressed ramets) and improves the growth of donor ramets (parent, older or unstressed ramets; Roiloa & Retuerto 2006; Wang *et al.* 2009; Roiloa *et al.* 2010; Xiao *et al.* 2011). Furthermore, the growth reduction of the recipient ramets is often equal to or greater than the growth increase of the donor ramets, so that fragmentation either does not affect (van Kleunen *et al.* 2000; Wang *et al.* 2009; Xiao *et al.* 2011) or decreases (Roiloa & Retuerto 2006; Roiloa *et al.* 2010) the growth of the whole clone. In the present study, we also found that clonal fragmentation increased biomass of parent ramets of *P. stratiotes* and decreased biomass of their offspring ramets. These results suggest that parent ramets of *P. stratiotes* export photosynthate to offspring ramets when they were connected (Alpert 1999; Roiloa & Retuerto 2012; Song *et al.* 2013), and that the connections between parent and offspring ramets confer higher costs to the parent ramets and higher benefits to the offspring ramets (de Kroon *et al.* 1996; Wang *et al.* 2009).

Surprisingly, however, we found that the biomass reduction of the offspring ramets was smaller than the biomass increase of the parent ramets, so that clonal fragmentation significantly increased biomass of the whole clone of *P. stratiotes*. The likely reason is that maintaining the stolon connections between the parent and offspring ramets incurred large costs (*e.g.* resource transportation and metabolic costs) to both the parent and offspring ramets (Oborny & Podani 1996; Alpert 1999; Wang *et al.* 2009). This exacerbated the growth reduction of the parent ramets caused by exporting photosynthate and reduced the growth increase of the offspring ramets caused by importing photosynthate.

Clonal fragmentation significantly decreased total biomass of offspring ramets, but did not affect the number of offspring ramets. As a result, fragmentation significantly decreased the size of individual offspring ramets. However, such effects were larger in the medium- and high-density treatments (with intraspecific neighbours) than in the low-density treatment (without neighbours), suggesting that the negative effects of fragmentation on individual offspring ramets are much more severe when these ramets face intense intraspecific competition.

Effects of fragmentation on intraspecific competition

Contrary to our expectation, we did not find a significant effect of density x fragmentation on growth of the whole clone of *P. stratiotes*, and fragmentation also did not significantly affect the log response ratio. These results suggest that clonal fragmentation did not affect intraspecific competition between clones of *P. stratiotes*. Previous studies also showed that clonal fragmentation had little effect on interspecific competition between clonal plants (Schmid & Bazzaz 1987; Amsberry *et al.* 2000; Pennings & Callaway 2000; Peltzer 2002) or intraspecific competition in *Solidago canadensis* (Schmid & Bazzaz 1987). However, clonal fragmentation was found to reduce competitive ability of *S. canadensis* when grown with *Poa pratense* and *Aster pilosus* (Hartnett & Bazzaz 1985). In the present study, clonal fragmentation did not significantly affect the number of offspring ramets of *P. stratiotes*, and changes in intraspecific competition intensity with time were similar with or without clonal fragmentation. Consequently, clonal fragmentation had little effect on intraspecific competition of *P. stratiotes*.

Because clonal fragmentation could greatly increase the growth of the whole clone of *P. stratiotes* but did not increase intraspecific competition, it may markedly contribute to its interspecific competitive ability and thus invasiveness (Pennings & Callaway 2000; Peltzer 2002). Further studies could be designed to test how clonal fragmentation affects competitive ability of floating plants when they face interspecific competitors, and how clonal fragmentation affects the growth and spread of floating plants when they invade local communities.

Effects of fragmentation on size–number relationships of offspring ramets

When clones of *P. stratiotes* were not fragmented, the size and number of offspring ramets was negatively correlated. This is because without fragmentation ramet number decreased but ramet size increased with increasing density. However, when clones of *P. stratiotes* were fragmented, both ramet size and number decreased with increasing density so that ramet size and number were positively correlated. Thus, the negative and positive size–number relationships of offspring ramets were caused by the different fragmentation-mediated responses of ramet number and size to density and thus to intraspecific competition. This result further indicates that whether selection favours the production of many small offspring ramets over a few large offspring ramets can be opposite, depending on the disturbance regime (Takada & Nakajima 1996; Sugiyama & Bazzaz 1998; Xie & Yu 2011). Production of fewer but bigger offspring ramets may be beneficial in dense and undisturbed environments (Stuefer *et al.* 2002), as we found in the high-density treatment without clonal fragmentation. With fragmentation, and thus lack of support from the parent ramets, production of fewer and smaller offspring ramets in the high-density environment is apparently a passive, density-mediated response (Takada & Nakajima 1996; Sugiyama & Bazzaz 1998; Xie & Yu 2011).

CONCLUSIONS

We conclude that clonal fragmentation cannot alter intraspecific competition between clones of *P. stratiotes*, but can increase biomass production of the whole clone. Thus, clonal fragmentation may contribute to the interspecific competitive ability and invasiveness, and intentional fragmentation should not be recommended as a measure to stop the rapid growth of this invasive species.

ACKNOWLEDGEMENTS

We thank Da-Yong Zhou and Gu Gong for assistance with measurements and harvests, and two anonymous reviewers for helpful comments. This research was supported by the Fundamental Research Funds for the Central Universities (BLYJ2 01307, TD-JC-2013-1) and the Program for New Century Excellent Talents in University (NECT-10-0234).

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Figure S1. Total number of offspring ramets per initial *Pistia stratiotes* ramet counted every other day.

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